Export to PDF: `python scripts/_convert_md_pdfs.py` (produces `scripts/Paper3_Methods_Plain.pdf` and `scripts/Paper3_Code_Design_Methodology_Map.pdf`).

Paper 3 — methods (single source of truth)

Literature and sensor papers: `Paper3_Github/Experiment papers/`. Analysis scripts: `scripts/H1_to_H3.R`, `scripts/H4_multivariate.R`, `scripts/Normalization_H1toH4.ipynb`.

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§1 Hypothesis

Confirmed wording (user, 2026-10-13; source of truth: `C:\Users\pandya\.gemini\GEMINI.md`, from `documents/Paper3_outline_ml.docx` for hypothesis text only):

- **H1:** Participants will show better physiological and perceived comfort in green than in impervious areas.
- **H2:** Physiological and perceived responses will be concordant (i.e., they correlate in direction).
- **H3:** Psychophysiological comfort will be more compromised during cycling than during walking.
- **H4:** The presence of noise and traffic will amplify thermal discomfort.

H4 in this repo: We test H4 as **within-bout** associations of environmental heat, noise, and co-exposures with continuous Empatica and eye-tracker signals, with **prespecified heat × sound** (and **MRT × sound**) interactions—i.e. whether louder traffic/noise **amplifies** the link between thermal stress and physiology, matching "noise and traffic amplify thermal discomfort." Questionnaires are not H4 outcomes (they support H1–H3).

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§2 Terms (glossary)

Term	Meaning
Participants	Eleven people in the main analysis file (default exclusions P2, P3, P5, P6, P7 apply only when explicitly requested).
Phases	Outdoor bouts: **BikeU**, **WalkU** (urban); **BikeG**, **WalkG** (green). Indoor, rest, and tram are outside H1–H3 bout contrasts; tram is excluded from the 8 min merge used here.
10 s epochs	Sensor streams aligned to **10 second** rows in the merged CSV. H4 models use these rows; H1–H3 use participant–phase aggregates.
8 min window	First **matched eight minutes** per participant–phase so every bout contributes the same duration. Input: `output/merged_all_11participants_8min.csv`.
Outcomes (H1–H3 & H4)	**26** physiology series: Empatica + eye-tracker (declared in `H4_multivariate.R`). Surveys (STAI, PRS, etc.) are for H1–H3 only.
Exposures (H4)	Environmental candidates from UCM, Atmotube, LYS, GVI, SVF—reduced to **16** representatives before fitting (§5).
Centering (H4)	Exposures: **within participant–phase** mean deviation, scaled by pooled SD of deviations (within-bout centering).
Random effects (H4)	`(1 | ParticipantID / PhaseID)` with within-phase residual correlation (AR(1) when supported). H1–H3: random intercept per participant on phase means.

Term	Meaning
Exertion	`empatica_vector_magnitude` forced in every H4 equation (movement adjustment).
Phase contrasts (H4 fixed effects)	Three **sum-to-zero** contrasts: WalkG, WalkU, BikeG vs grand mean (BikeU derived for reporting).

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§3 What we do (pipeline)

Upstream colleague scripts in `Paper3_Github/` merge wearables, UCM weather and **dBA** sound, bilateral Atmotube PM, LYS light, GVI/SVF, and questionnaires into `output/merged_all_11participants.csv`; this workspace trims to the **8 min** file above. **`scripts/Normalization_H1toH4.ipynb`** applies the shared transform rule (natural log on seven HRV measures only). **`scripts/H1_to_H3.R`** tests H1–H3 on phase-level models. **`scripts/H4_multivariate.R`** runs the H4 selection funnel (Steps 1–4), then fits **26 outcomes × 6 exposure equations** at 10 s (Step 5 / H4.1). No LASSO, stepwise, or outcome-driven dropping of exposures.

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§4 H1–H3 (three bullets each)

H1 — Green vs impervious (urban)

- **Question:** On the same travel mode, does a green corridor differ from an urban corridor on physiology and perceived comfort?
- **Test:** Pre-planned **Green minus Urban** contrasts within bike and within walk, from mixed models on each person's phase means (26 physiology + survey outcomes where applicable; random intercept per person).
- **Why:** Phase ID encodes route and mode; contrasts were fixed before viewing results.

H2 — Physio–psycho concordance

- **Question:** When someone feels more comfortable or restored on a bout, do body and gaze signals move in the **same direction** as those ratings?
- **Test:** Repeated-measures correlation between questionnaires and physiological measures at participant–phase level (multiple pairs, one Benjamini–Hochberg family).
- **Why:** Estimates **within-person** alignment, not between-route differences.

H3 — Cycling vs walking

- **Question:** Is psychophysiological comfort **more compromised** when **cycling** than when **walking** (within urban and within green)?
- **Test: Walk minus Bike** contrasts (WalkU vs BikeU, WalkG vs BikeG) from the same phase-mean models as H1.
- **Why:** Mode and environment are crossed; these contrasts change mode while holding route type fixed.

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§5 H4 — selection and equations (tied to §1)

Hypothesis link: H4 predicts that **noise and traffic amplify thermal discomfort**. Step 5 therefore includes, for every fitted equation, **each heat term × `ucm_SND_dBA`** plus **`ucm_MRT × sound`**, alongside thermal and co-exposure main effects—so amplification by sound is explicit, not post hoc.

Steps 1–5 (written before viewing fit results)

Step	What happens	Count (typical)
1 — Inventory	Classify every merged column: outcome Y, exposure X, movement Z, metadata/QC, or out of scope. Outcomes **never** enter the exposure funnel.	230 → **124** advance

Step	What happens	Count (typical)
2 — Representatives	Fixed consolidation (bilateral Atmo/LYS averages, one wind/sound/GVI/SVF choice, drop derived/alternate formulations). **No** outcome-based dropping.	124 → **16**
3 — Grouping	Within-person correlation on phase-mean complete cells; **no** VIF elimination. All Step-2 eligibles advance.	**16 → 16**
4 — Equation designs	Six **pre-specified** layouts: shared non-thermal covariates + **one** heat representation + **one** PM size per equation. Rank / grouped GVIF checks only—**no** automatic term deletion.	**6** designs
5 — H4.1 fit	10 s mixed models: **26 outcomes × 6 equations**. Centered exposures, phase contrasts, exertion, interaction rule below. Cluster-robust inference (11 participants).	156 model families

16 retained exposures (after Step 2): `ucm_AIR_temp`, `ucm_AIR_RH`, `ucm_WIND_AWS`, `ucm_HUMIDEX`, `ucm_PET`, `ucm_UTCI`, `ucm_MRT`, `ucm_SUN_Gh`, `ucm_SND_dBA`, `atmo_pm1_avg`, `atmo_pm2.5_avg`, `atmo_pm10_avg`, `lys_lux_avg`, `lys_kelvin_avg`, `gvi_50m`, `SVF_Holmer`.

Shared covariate block in every equation: all 16 **except** the equation's chosen PM fraction, plus forced **`empatica_vector_magnitude`**, three phase sum-to-zero terms, and the interaction terms below.

Six equations

Eq	Role	Heat representation	PM term
1	**Primary**	UTCI	PM2.5 (`atmo_pm2.5_avg`)
2	Heat sensitivity	Air temp + RH + apparent wind (`ucm_WIND_AWS`)	PM2.5
3	Heat sensitivity	HUMIDEX	PM2.5
4	Heat sensitivity	PET	PM2.5
5	PM-size sensitivity	UTCI	PM1
6	PM-size sensitivity	UTCI	PM10

Readable formula pattern (each equation):

```
outcome ~ [heat terms for this eq] + [non-thermal covariates except this eq's PM] + [this eq's PM] + phase contrasts + exertion + (heat terms × sound) + (MRT × sound) + random effects
```

Interaction rule (H4 confirmatory structure)

- For **each heat term in that equation** (UTCI alone; or HUMIDEX; or PET; or air temp + RH + wind): include **heat × `ucm_SND_dBA`**.
- **`ucm_MRT`:** always **`MRT × sound`**, even when MRT is not the designated heat index for that equation.
- **All other covariates:** main effects only (no lux×sound, PM×sound, etc.).

FDR (Step 5 reporting)

Benjamini–Hochberg on **thermal main effects** only: family = one thermal term × one equation × **26** outcomes. Heat×sound and MRT×sound are **prespecified** and reported; they are **not** the primary FDR family.

Short meeting summary (formerly "one page")

Funnel figure: `output/statistical_analysis/results/H4_results/H4_Selection_Flow.pdf`. Primary narrative equation: **Eq 1 (UTCI + PM2.5)**. Coefficients: `_work/H4_models/H4_coefficients.csv`. Narrative results: `H4_Report.md`.

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§6 Proof files (audit trail)

Use **only** these for "show your work" on H4 column and design decisions:

- `output/statistical_analysis/results/H4_results/_work/H4_Multivariate_Filtration_Results.csv` — Step 1–3 register (every column decision).
- `output/statistical_analysis/results/H4_results/H4_Selection_Flow.pdf` — selection funnel figure (Steps 1–4).

Regenerate the figure only: `H4_RENDER_FLOW_ONLY=1 Rscript scripts/H4_multivariate.R`.

Run full analysis: `Rscript scripts/H1_to_H3.R` then `Rscript scripts/H4_multivariate.R`.

What we do not do: LASSO, stepwise, outcome-driven exposure deletion, questionnaire outcomes inside H4, or auto-dropping predictors when VIF is high (flag/block instead).